Tarea 0

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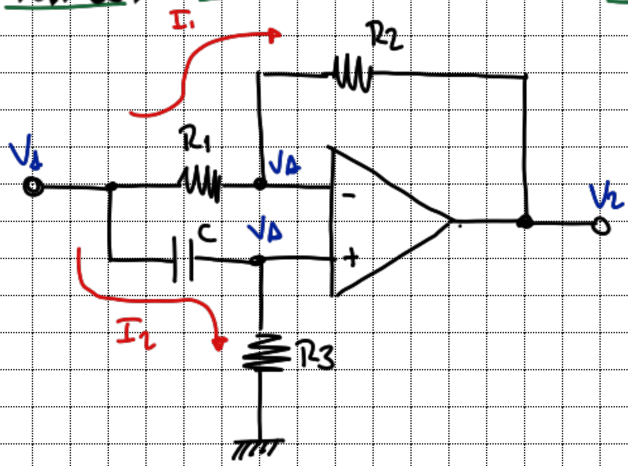


$$\frac{V_1 - V_A}{R_1} = I_1 \qquad I_1 = \frac{V_A - V_2}{R_2} \quad (2)$$

$$\frac{V_1 - V_A}{\frac{1}{SC}} = I_2 \qquad I_2 = \frac{V_A}{R_3} \quad (1)$$

(1)

$$(V_1 - V_A)SC = \frac{V_A}{R_3}$$

$$V_1 SC = V_A\left(\frac{1}{R_3} + SC\right)$$

$$V_A = \frac{SC}{\frac{1}{R_3} + SC} V_1$$

$$V_A = \frac{S\cancel{C}}{\cancel{C}\left(S + \frac{1}{R_3C}\right)} V_1$$

$$\boxed{V_A = \frac{S}{S + \frac{1}{R_3C}} V_1}$$

(2)

$$\frac{V_1 - V_A}{R_1} = \frac{V_A - V_2}{R_2}$$

$$V_A\left(\frac{1}{R_1} + \frac{1}{R_2}\right) = \frac{V_1}{R_1} + \frac{V_2}{R_2}$$

$$V_A\left(\frac{R_2 + R_1}{R_1 R_2}\right) = \frac{V_1}{R_1} + \frac{V_2}{R_2}$$

$$V_1\left(\frac{S}{S + \frac{1}{R_3C}}\right)\left(\frac{R_2 + R_1}{R_1 R_2}\right) = \frac{V_1}{R_1} + \frac{V_2}{R_2}$$

$$V_1\left(\left(\frac{S}{S + \frac{1}{R_3C}}\right)\left(\frac{R_2 + R_1}{R_1 R_2}\right) - \frac{1}{R_1}\right) = \frac{V_2}{R_2}$$

$$\frac{V_2}{V_1} = R_2\left(-\frac{1}{R_1} + \left(\frac{S}{S + \frac{1}{R_3C}}\right)\left(\frac{R_2 + R_1}{R_1 R_2}\right)\right)$$

$$\frac{V_2}{V_1} = Av = -\frac{R_2}{R_1} + \left(\frac{S}{S + \frac{1}{R_3C}}\right)\left(\frac{R_2 + R_1}{R_1}\right)$$

$$A_v = -\frac{R_2}{R_1} + \left(\frac{s}{s+\frac{1}{R_3C}}\right)\left(\frac{R_2+R_1}{R_1}\right)$$

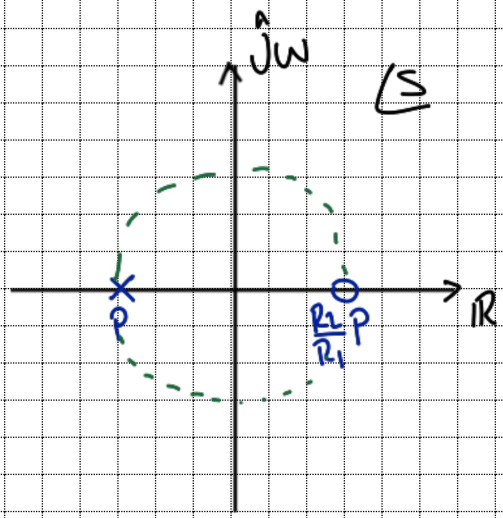
$$-\frac{R_2}{R_1} + \frac{sR_3C}{sR_3C+1}\cdot\frac{R_2}{R_1} + \frac{sR_3C}{sR_3C+1}$$

$$-\frac{R_2\,(sR_3C+1)}{R_1\,(sR_3C+1)} + \frac{R_2\,sR_3C}{R_1\,(sR_3C+1)} + \frac{sR_3C}{sR_3C+1}$$

$$\frac{\cancel{-sR_2R_3C} - R_2 + \cancel{R_2sR_3C}}{R_1\,(sR_3C+1)} + \frac{sR_3C}{sR_3C+1}\,\frac{R_1}{R_1}$$

$$A_v = \frac{sR_3CR_1 - R_2}{R_1\,(sR_3C+1)} = \frac{sR_3CR_1 - R_2}{sR_3R_1C + R_1}$$

$$A_v = \frac{R_3\cancel{C}R_1}{R_3\cancel{C}R_1}\;\frac{s - \frac{R_2}{R_1CR_3}}{s + \frac{\cancel{R_1}}{\cancel{R_1}CR_3}}$$



$$\boxed{A_v = \frac{s - \frac{R_2}{R_1CR_3}}{s + \frac{1}{CR_3}}}$$

$$\frac{1}{CR_3} = P$$

$s = jw$

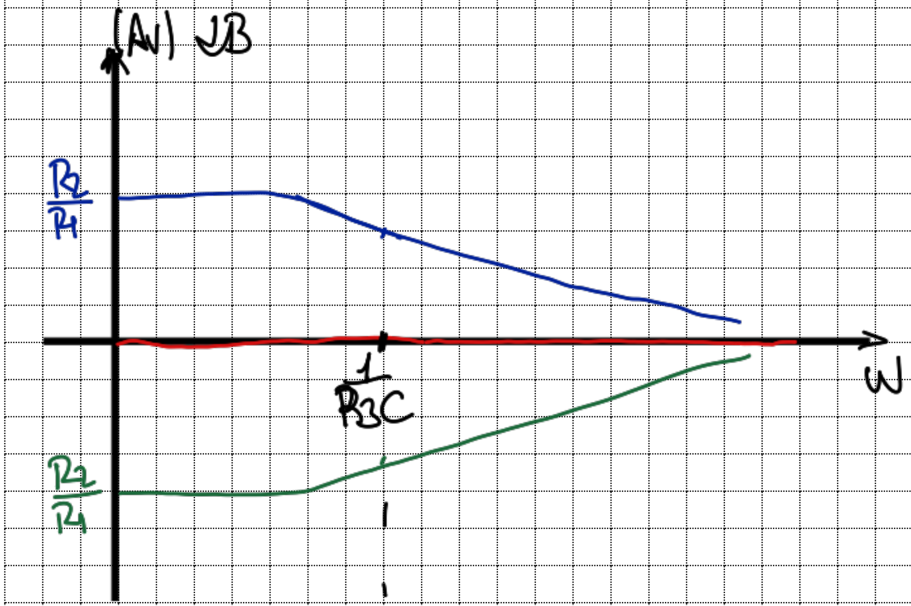
$$A_v(s=jw) = \frac{jw - \frac{R_2}{R_1CR_3}}{jw + \frac{1}{CR_3}}$$

$$|A_v(s=jw)| = \frac{\sqrt{\left(\frac{R_2}{R_1CR_3}\right)^2 + w^2}}{\sqrt{\left(\frac{1}{CR_3}\right)^2 + w^2}}$$

• Si $w=0 \Rightarrow |A_v| = \dfrac{\frac{R_2}{R_1\cancel{CR_3}}}{\frac{1}{\cancel{CR_3}}} = \dfrac{R_2}{R_1}$

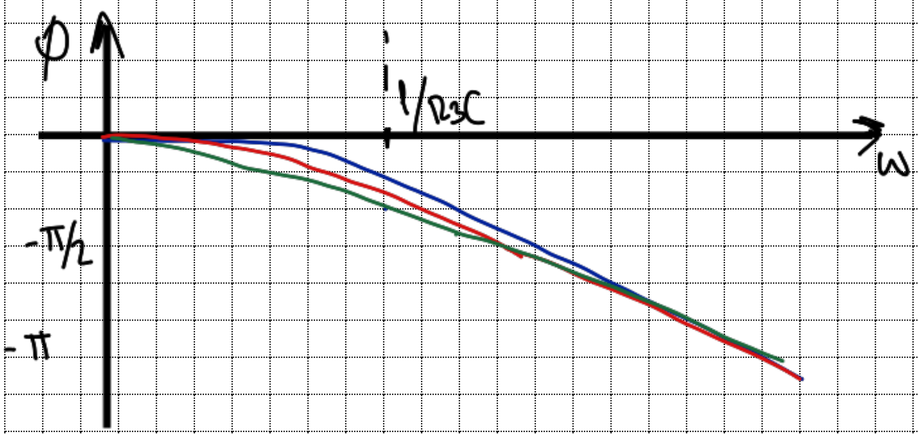
- Si $\omega \to \infty$

$$\lim_{\omega\to\infty} = \frac{\sqrt{\left(\frac{R_2}{R_1 C R_3}\right)^2 + \omega^2}}{\sqrt{\left(\frac{1}{C R_3}\right)^2 + \omega^2}} \simeq \lim_{\omega\to\infty} \frac{\sqrt{\omega^2}}{\sqrt{\omega^2}} = \lim_{\omega\to\infty} \frac{|\omega|}{|\omega|} = 1$$



- Si $R_2 > R_1$
- Si $R_2 = R_1$
- Si $R_2 < R_1$

$$\phi = tg^{-1}\left(\frac{\omega}{-\frac{R_2}{R_1}\frac{1}{CR_3}}\right) - tg^{-1}\left(\frac{\omega}{\frac{1}{CR_3}}\right)$$



$$\boxed{\lim_{\omega\to 0} \phi = 0}$$

$$\lim_{\omega\to\infty} \underbrace{tg^{-1}\left(-\frac{R_1R_3C\omega}{R_2}\right)}_{\to -\pi/2} - \underbrace{tg^{-1}(CR_3\omega)}_{\to \pi/2}$$

$$\boxed{\lim_{\omega\to\infty} \phi = -\pi}$$

(2) $$A_V(s) = \left(\frac{sR_3CR_1 - R_2}{sR_3R_1C + R_1}\right) \frac{1/R_1}{1/R_1}$$

$$R_2' = R_2/R_1 \qquad C' = C/R_1$$

$$A_V(s) = \frac{sR_3R_1C' - R_2'}{sR_3R_1C' + 1}$$

$$\$ = R_3R_1C'$$

$$\boxed{A_V(\$) = \frac{\$ - R_2'}{\$ + 1}}$$

Si $R_2 = R_1$ "Pasatodo"
$R_2 > R_1$ "Pasa-bajos"
$R_2 < R_1$ "Pasa-altos"